Exp 9 MATLAB CODE

```
clc;
clear;
close all;

Fs = 100000;
t = 0:1/Fs:0.02;

Am = 1;
fm = 1000;
fc = 10000;
mu = 0.7;

% Message Signal
m = Am*cos(2*pi*fm*t);

% AM Signal Generation
s = (1 + mu*m).*cos(2*pi*fc*t);

% Add AWGN Noise
SNR = 10;
noisy_signal = awgn(s,SNR,'measured');

% AM Detection using Envelope Detector
envelope = abs(hilbert(noisy_signal));
detected = (envelope - mean(envelope))/mu;

% Low-pass Filter
```

```
[b,a] = butter(5,2000/(Fs/2),'low');
recovered = filtfilt(b,a,detected);

% Calculate Mean Squared Error
MSE = mean((m-recovered).^2);

% Plot Results
figure;

subplot(4,1,1);
plot(t,m);
title('Original Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,s);
title('Generated AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,3);
plot(t,noisy_signal);
title('AM Signal with Noise');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;


subplot(4,1,4);
plot(t,recovered);
title('Recovered Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;


fprintf('SNR = %d dB\n',SNR);
fprintf('Mean Squared Error = %.6f\n',MSE);
```

